Problem 12.2

At a certain interest rate, the present values of the following two payment patterns are equal:

1. 200 at the end of 5 years plus 500 at the end of 10 years;
2. 400.94 at the end of 5 years.

At the same interest rate, 100 invested now plus 120 invested at the end of 5 years will accumulate to P at the end of 10 years. Calculate P.

Solution

Let i be the annual effective interest rate. From the equality of the present values,

$$\frac{400.94}{(1+i)^5} = \frac{200}{(1+i)^5} + \frac{500}{(1+i)^{10}},$$
$$400.94(1+i)^5 = 200(1+i)^5 + 500.$$

Set $x = (1+i)^5$. Then

$$200.94x = 500,$$
$$x = \frac{500}{200.94} \approx 2.488304967.$$

At year 10, the 100 invested at time 0 has accumulated by a factor $(1+i)^{10} = x^2$, and the 120 invested at year 5 has accumulated by a factor $(1+i)^5 = x$. Therefore,

$$\begin{aligned} P &= 100x^2 + 120x \\ &= 100(2.488304967)^2 + 120(2.488304967) \\ &\approx 917.762757. \end{aligned}$$

Thus, the accumulated amount at the end of 10 years is

$$\boxed{P \approx 917.76}.$$